Redline: AI Resume Analyser

Windows setup guide for beginners

Step 1. Software you need

Python 3.10, 3.11 or 3.12 (64-bit) and **VS Code**. Git is optional. **Node.js is NOT needed**: the frontend is plain HTML/CSS/JS served by Flask. No model or dataset download is needed either. Internet is needed once to install packages and to load web fonts.

Step 2. Install the software

Python: download from python.org/downloads, run the installer and **tick 'Add python.exe to PATH'** before clicking Install Now. VS Code: download from code.visualstudio.com and install with defaults. Then open a new Command Prompt and check:

```
py --version
```

Step 3. Open the project in VS Code

Unzip the project, e.g. to C:\Projects\ai-resume-analyser. In VS Code choose File > Open Folder and pick that folder. Open the terminal with Ctrl + ` and make it Command Prompt (the down-arrow next to + > Command Prompt).

Step 4. Create and activate the virtual environment

Run these in the VS Code terminal, one line at a time. The prompt should start with (venv).

```
cd backend
py -m venv venv
venv\Scripts\activate
```

Step 5. Install Python dependencies

This installs Flask, scikit-learn, pypdf, python-docx, python-dotenv and waitress. It takes a few minutes.

```
pip install -r requirements.txt
```

Step 6. Frontend dependencies

None. Skip this step: there is no npm install.

Step 7. Configure .env

Copy the example file. The defaults work as they are; no API keys are used.

```
copy .env.example .env
```

Step 8. ML model / dataset

Nothing to place. The ML part (TF-IDF + cosine similarity) is computed live from the resume and the job description each time.

Steps 9 to 11. Start the app and open it

Make sure (venv) is shown, then run the command below. One server serves both the API and the website. Leave the terminal open and visit **http://localhost:5000** in Chrome.

```
python app.py
```

Step 12. Test with the sample resume

Click the upload box, choose **sample/sample_resume.pdf**, click 'Use a sample' above the job description box, then press Analyse resume. You should see a score of about 89/100 and the missing skill 'nlp'. Test errors too: upload a .txt file (unsupported type message) or a renamed file (invalid document message).

Step 13. Common errors and fixes

'py' is not recognized: reinstall Python with Add to PATH ticked, then reopen VS Code.
Running scripts is disabled (PowerShell): use Command Prompt, or run the first line below.
ModuleNotFoundError: the venv is not active; run venv\Scripts\activate then pip install -r requirements.txt.
Address already in use: edit .env, set PORT=5001, restart and open localhost:5001.
pip install fails on scikit-learn: use Python 3.10-3.12 (not 3.13+ or 32-bit), then run: python -m pip install --upgrade pip.
'Cannot reach the server' in the browser: the backend is not running; run python app.py again.
'No readable text found': the PDF is a scanned image; use a text-based PDF or DOCX.

```
Set-ExecutionPolicy -Scope Process Bypass
```

Step 14. Stop the servers

Click in the terminal and press Ctrl + C. Close the terminal when done.

Step 15. Restart later

Open the folder in VS Code and run:

```
cd backend
venv\Scripts\activate
python app.py
```

Step 16. Production-style run

Waitress is already installed from requirements.txt. Run this from the backend folder and open http://localhost:5000. There is no frontend build step.

```
waitress-serve --port=5000 app:app
```